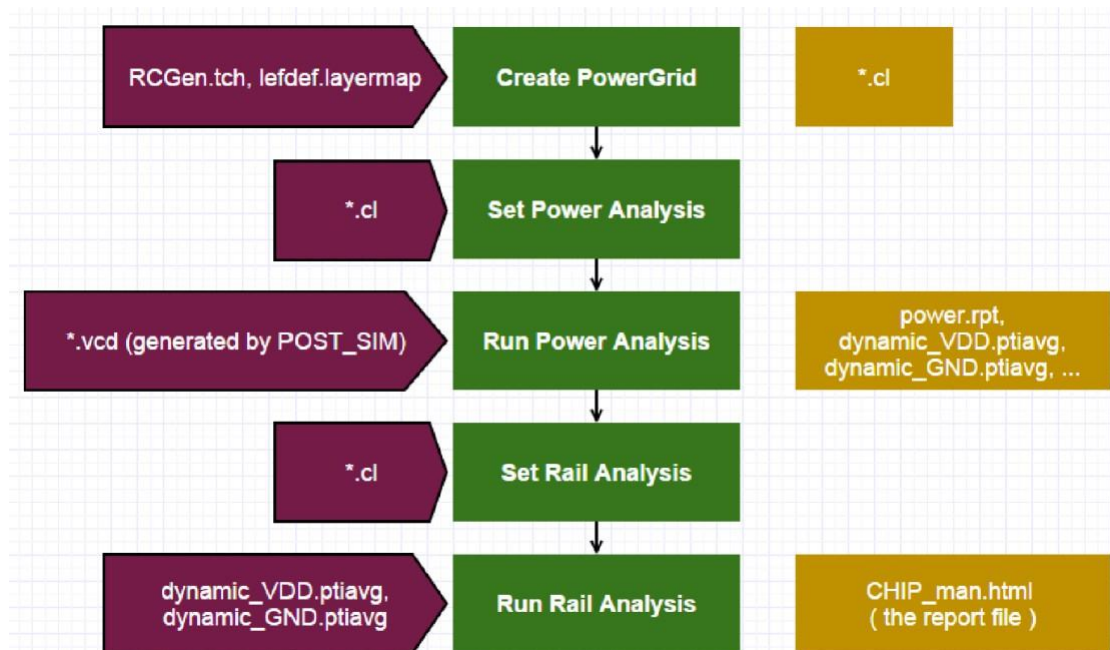


NYCU-EE IC Design LAB - Fall 2025

Lab12 Power Rail Analysis Practice Tutorial

1. Flow overview

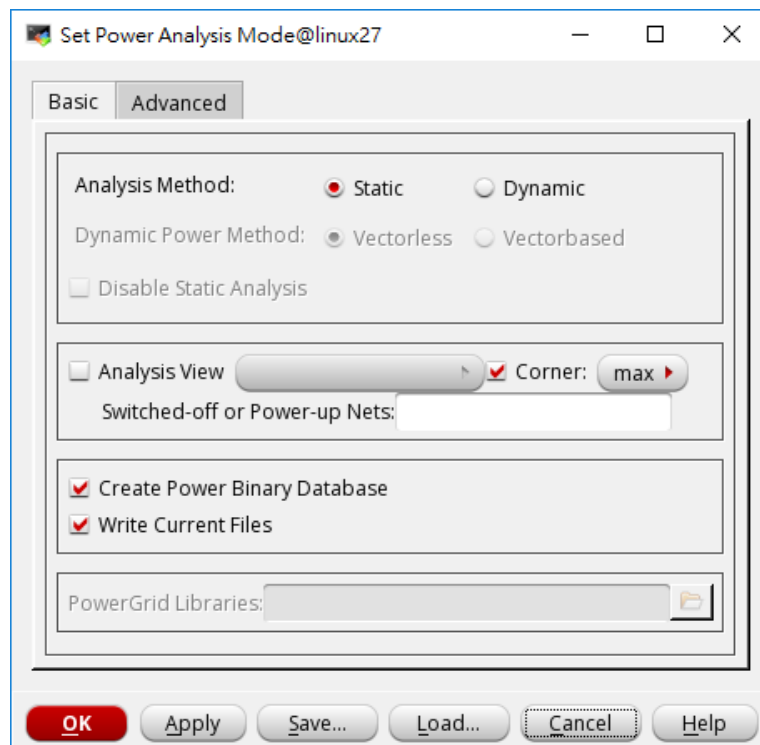
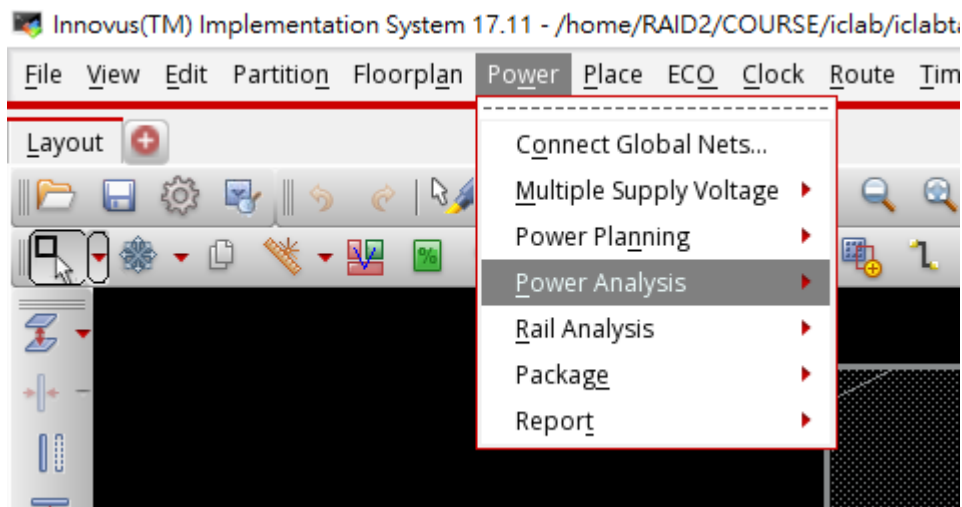


2. Set environment

- `unix% cd Lab12_2025S/Exercise/05_APR`
- `unix% mkdir power_log` (You will save all the things here)
- `unix% innovus`
- Restore the design **DBS/CHIP.inn**

3. Static Power Analysis

1. Save CHIP.v
2. Write CHIP.sdf
3. Run post simulation at 06_POST, the generated waveform CHIP_POST.fsdb will be used for power rail analysis.
4. In the innovus menu, open **Power -> Power Analysis -> Setup**



- i. Click **OK**

5. In the innovus menu, open **Power -> Power Analysis -> Run**

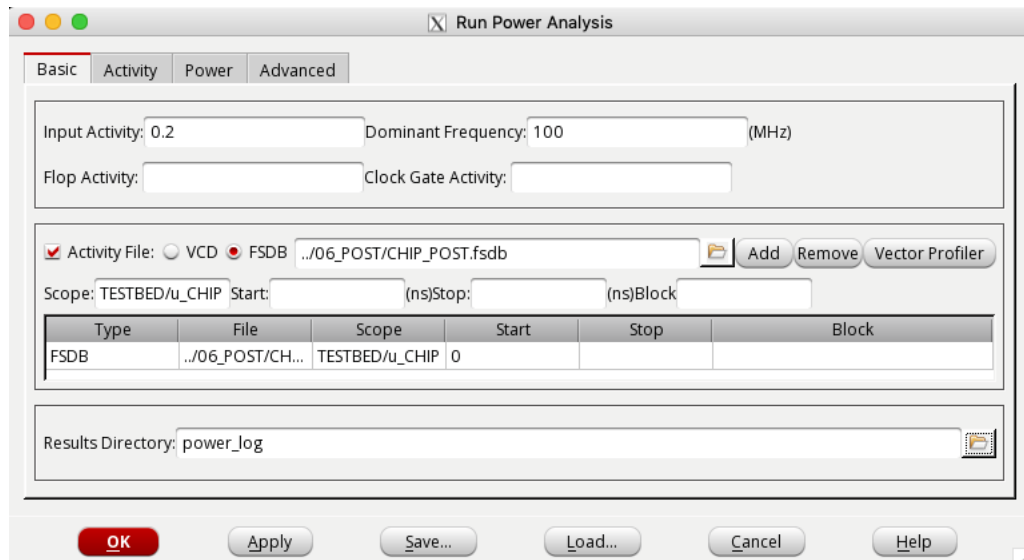
i. ♦Activity FILE ♦FSDB

ii. Fill the information:

- Select CHIP_POST.fsdb (from 06_POST)
- Scope: TESTBED/u_CHIP
- Press **Add**

iii. Results Directory: power_log

iv. Click **OK**.

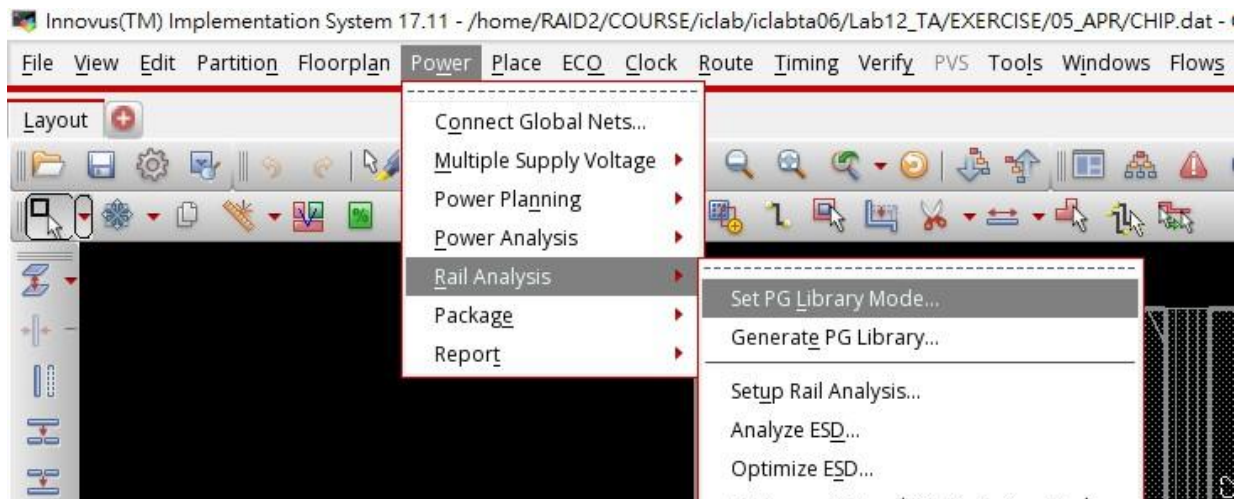


v. Results appear at terminal

```
Total Power
-----
Total Internal Power:      56.34742919      89.3507%
Total Switching Power:     6.69368832      10.6142%
Total Leakage Power:       0.02211333      0.0351%
Total Power:               63.06323143
```

4. Create Power Grid Library

1. In the innovus menu, open **Power -> Rail Analysis -> Set PG Library Mode**



- i. Cell type: ♦Tech Only
- ii. Filler Cell Names: **FILL1 FILL16 FILL2 FILL32 FILL4 FILL64 FILL8**
- iii. Extraction
 - Extraction tech file: **05_APR/RC/icecaps.tch** (File of type: All files(*))
 - LEF Layermap: **05_APR/layermap/lefdef.layermap.libgen**(File of type: All files(*))
- iv. Supply Pins
 - Voltages: **1.8**
 - Power pin: **VCC**
 - Ground pin: **GND**
- v. Click

Basic Advanced

Cell Type ☒ Tech Only ☐ Std Cells ☐ Macros/IO

Filler Cell Names : FILL4 FILL64 FILL8 Decap Cell Names

Capacitance ☒ Area Cap(fF/um^2) 0.5

☐ Simulation

Spice subckt netlist

Spice model

XYScaling

Corners

Add Remove

Spice Model	Spice Corner(s)

Extraction

Extraction tech file RC/icecaps.tch

GDS Files

☐ GDS port generation

LEF Layermap ef.layermap.libgen

GDS Layermap

Supply Pins

Power Pins VCC Voltages 1.8

Ground Pins GND

OK Apply Save... Load... Cancel Help

2. In the innovus menu, open **Power -> Rail Analysis -> Generate PG Library**

- i. Choose power_log
- ii. Click **OK**

Generate PG Library

Result Directory power_log

Library Prefix

OK Apply Cancel Help

- iii. Check if the directory **techonly.cl** exists (under power_log/)

3. In the innovus menu, open **Power -> Rail Analysis -> Set PG Library Mode**

- i. Cell type: ♦Std Cells
- ii. Filler Cell Names: **FILL1 FILL16 FILL2 FILL32 FILL4 FILL64 FILL8**
- iii. Extraction
 - Extraction tech file: **05_APR/RC/icecaps.tch**
 - LEF Layermap: **05_APR /layermap/ lefdef.layermap.libgen**
- iv. Supply Pins
 - Voltage: **1.8**
 - Power pin: **VCC**
 - Ground pin: **GND**
- v. Click **OK**.

The screenshot shows the 'Set PG Library Mode' dialog box with the 'Basic' tab selected. The 'Cell Type' section has 'Std Cells' selected. The 'Filler Cell Names' field contains 'FILL4 FILL64 FILL8'. The 'Capacitance' section has 'Area Cap(fF/um^2)' checked with a value of 0.5. The 'Extraction' section has 'Extraction tech file' set to 'RC/icecaps.tch' and 'LEF Layermap' set to 'ef.layermap.libgen'. The 'Supply Pins' section has 'Power Pins' set to 'VCC' with a voltage of 1.8, and 'Ground Pins' set to 'GND'. The 'OK' button is highlighted in red.

Basic	
Cell Type	☐ Tech Only ☒ Std Cells ☐ Macros/IO
Filler Cell Names	FILL4 FILL64 FILL8
Decap Cell Names	
Capacitance	☒ Area Cap(fF/um^2) 0.5
	☐ Simulation
Spice subckt netlist	
Spice model	
XYScaling	
Corners	
Add Remove	
Spice Model	
Spice Corner(s)	
Extraction	Extraction tech file RC/icecaps.tch
	LEF Layermap ef.layermap.libgen
	GDS Files
	GDS Layermap
	☐ GDS port generation
Supply Pins	Power Pins VCC
	Voltagess 1.8
	Ground Pins GND
OK Apply Save... Load... Cancel Help	

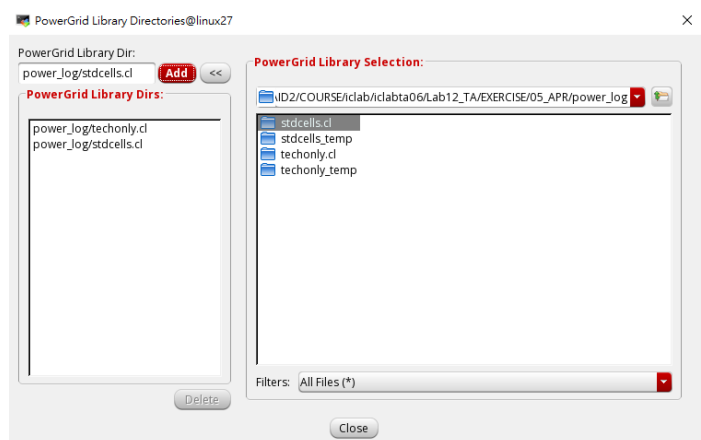
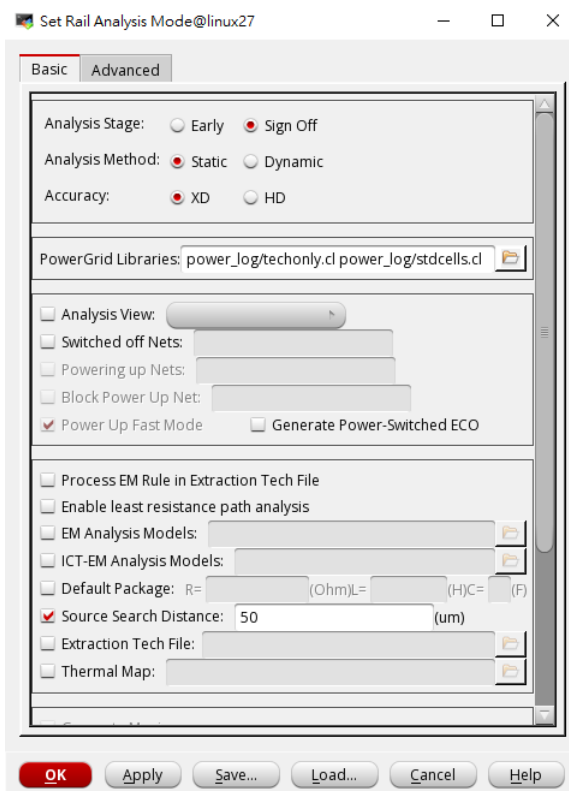
4. In the innovus menu, open **Power -> Rail Analysis -> Generate PG Library**

- i. Click **OK**
- ii. Check if the directory **stdcells.cl** exists (under power_log/)

5. Rail Analysis

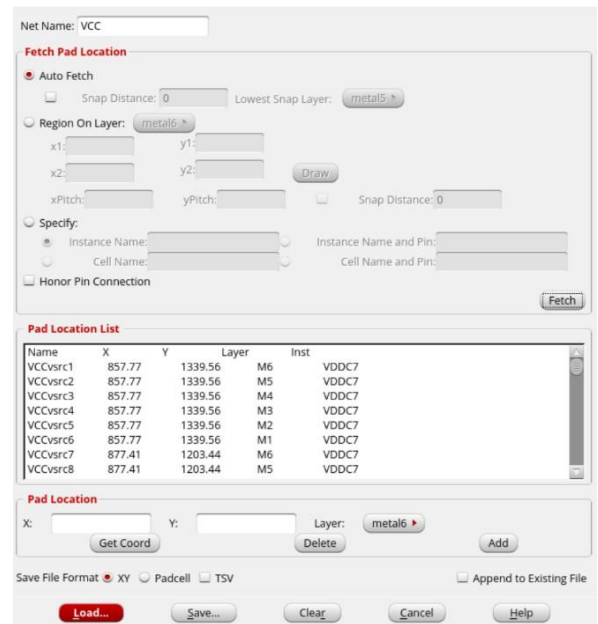
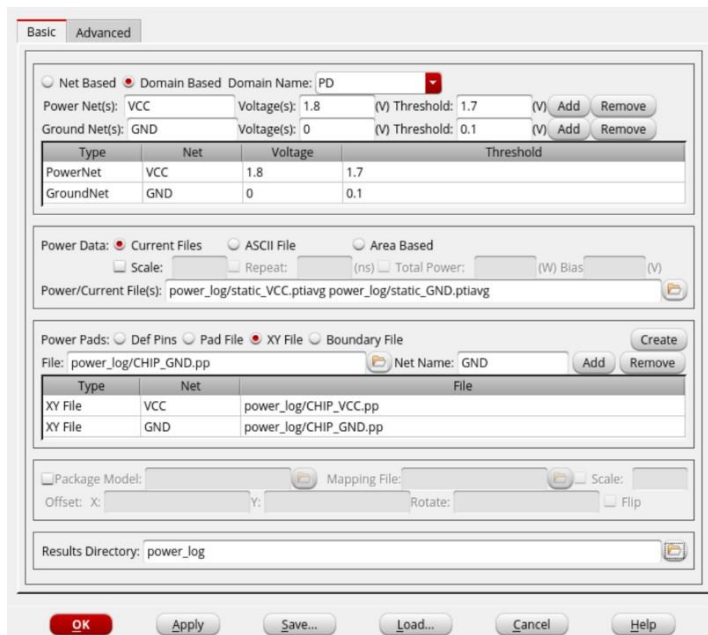
1. In the innovus menu, open **Power -> Rail Analysis -> Setup Rail Analysis**

- i. Analysis Method: **Static**
- ii. PowerGrid Libraries:
power_log/**technoly.cl** (should be added first)
power_log/**stdcells.cl**
- iii. Click **OK**



2. In the innovus menu, open **Power -> Rail Analysis -> Run Rail Analysis**

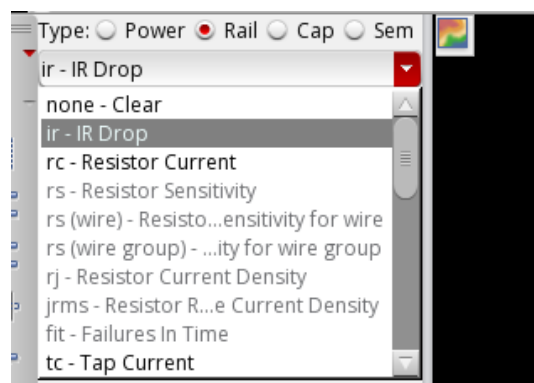
- i. ◆ Domain Based Domain Name: PD
- ii. Power Net(s): VCC Voltage(s): 1.8 Threshold:1.7 (Press **ADD**)
Power Net(s): GND Voltage(s): 0 Threshold:0.1 (Press **ADD**)
($1.8v * 5\% \approx 0.1v$)
- iii. Power/Current Files(s):
power_log/static_VCC.ptiavg
power_log/static_GND.ptiavg
- iv. Power Pads: ◆ XY File
- v. Click **Create**
 - Net Name: VCC
 - Click **Fetch**
 - Save as **power_log/CHIP_VCC.pp**
- vi. Click **Create** again
 - Net Name: GND
 - Click **Fetch**
 - Save as **power_log/CHIP_GND.pp**
- vii. Click **Cancel**
- viii. File: **power_log/CHIP_VCC.pp** Net Name: VCC (press **ADD**)
- ix. File: **power_log/CHIP_GND.pp** Net Name: GND (press **ADD**)
- x. Results Directory: **power_log**
- xi. Click **OK**

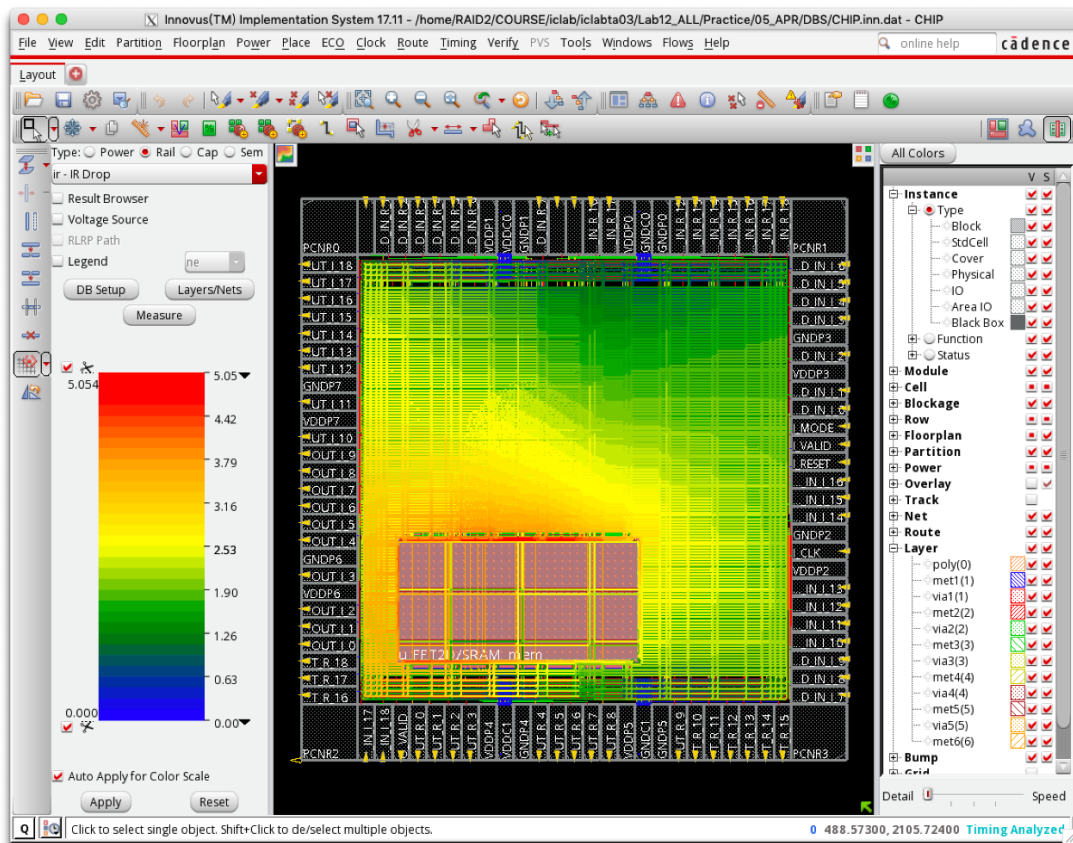


6. Power & IR Drop Results

1. In the innovus menu, open **Power -> Report -> Power & Rail Result**

- i. ◆ Auto Apply for Color Scale
- ii. Click **DB Setup**
 - Power Databases: power_log/power.db
 - Rail Database: power_log/**PD_25C_avg_1**
 - Click **OK**
- iii. Type: ◆ Rail
- iv. Choose **ir – IR Drop**





i. ♦Result Browser

The following shows the distribution of IR Drop (they should in the range **0.1V**)

